

K8s-9

1. Execute all the steps shown in video.

Deploy a NGINX application

- Create a deployment yaml file

```
root@Master:~# vi deployment.yaml
root@Master:~#
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

- Now apply it and check created or not nginx

```
root@Master:~# kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
root@Master:~# kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	2/2	2	2	14s
recreate-deployment	1/1	1	1	20d
surge-deployment	1/1	1	1	20d

- Now check pods

```
root@Master:~# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-7769f8f85b-tlv2m	1/1	Running	0	23s
nginx-deployment-7769f8f85b-z22sm	1/1	Running	0	23s
recreate-deployment-7c6d446b89-k2jvd	1/1	Running	1 (4m52s ago)	14h
surge-deployment-c95fcdf96-kmsmn	1/1	Running	1 (4m52s ago)	14h

```
root@Master:~#
```

Expose it using a Node Port Service

- Now created Node port service yaml file

```
root@Master:~# vi service.yaml
root@Master:~#
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30001
~
```

- Apply and get Svc

```
root@Master:~# vi service.yaml
root@Master:~#
root@Master:~# kubectl apply -f service.yaml
service/nginx-service configured
root@Master:~# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
apache-service	ClusterIP	10.102.172.16	<none>	80/TCP	24d
firstpod-service	ClusterIP	10.99.160.6	<none>	80/TCP	24d
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	25d
nginx-nodeport-service	NodePort	10.110.94.8	<none>	80:30080/TCP	24d
nginx-rc-service	NodePort	10.103.111.115	<none>	80:30081/TCP	24d
nginx-service	NodePort	10.104.0.105	<none>	80:30001/TCP	19d

```
root@Master:~#
```

- Test from inside the cluster

```
root@Master:~# kubectl run test --image=busybox --rm -it --restart=Never -- sh
If you don't see a command prompt, try pressing enter.
/ # kubectl run test --image=busybox --rm -it --restart=Never -- sh
sh: kubectl: not found
```

- To check Endpoints

```
sh: kubectl: not found
/ # exit
pod "test" deleted
pod default/test terminated (Error)
root@Master:~# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
apache-service	ClusterIP	10.102.172.16	<none>	80/TCP	24d
firstpod-service	ClusterIP	10.99.160.6	<none>	80/TCP	24d
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	25d
nginx-nodeport-service	NodePort	10.110.94.8	<none>	80:30080/TCP	24d
nginx-rc-service	NodePort	10.103.111.115	<none>	80:30081/TCP	24d
nginx-service	NodePort	10.104.0.105	<none>	80:30001/TCP	19d

```
root@Master:~# kubectl get endpoints nginx-service
```

NAME	ENDPOINTS	AGE
nginx-service	192.168.219.101:80,192.168.219.87:80	19d

- To install ingress

```
root@Master:~# kubectl apply -f https://raw.githubusercontent.com/kubernetes/ingress-nginx/main/deploy/static/provider/cloud/deploy.yaml
namespace/ingress-nginx created
serviceaccount/ingress-nginx created
serviceaccount/ingress-nginx-admission created
role.rbac.authorization.k8s.io/ingress-nginx created
role.rbac.authorization.k8s.io/ingress-nginx-admission created
clusterrole.rbac.authorization.k8s.io/ingress-nginx created
clusterrole.rbac.authorization.k8s.io/ingress-nginx-admission created
rolebinding.rbac.authorization.k8s.io/ingress-nginx created
rolebinding.rbac.authorization.k8s.io/ingress-nginx-admission created
clusterrolebinding.rbac.authorization.k8s.io/ingress-nginx created
```

- check svc and pods

```
ingressclass.networking.k8s.io/nginx created
validatingwebhookconfiguration.admissionregistration.k8s.io/ingress-nginx-admission created
root@Master:~# kubectl get pods -n ingress-nginx
NAME                                READY    STATUS    RESTARTS   AGE
ingress-nginx-controller-7c87c8df67-bvcm6  0/1      Running   0           12s
root@Master:~# kubectl get svc -n ingress-nginx
NAME                                TYPE          CLUSTER-IP      EXTERNAL-IP    PORT(S)                                AGE
ingress-nginx-controller            LoadBalancer  10.106.255.122   <pending>      80:31480/TCP,443:32553/TCP           23s
ingress-nginx-controller-admission  ClusterIP     10.101.100.172   <none>         443/TCP                               23s
root@Master:~# ^C
root@Master:~# kubectl get pods -n ingress-nginx
NAME                                READY    STATUS    RESTARTS   AGE
ingress-nginx-controller-7c87c8df67-bvcm6  1/1      Running   0           78s
```

- Create the Ingress resource
- Create ingress yaml file

```
root@Master:~# vi ingress.yaml
root@Master:~#
```

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-ingress
spec:
  ingressClassName: nginx
  rules:
  - host: teki.saber.com
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: nginx-service
            port:
              number: 80
```

- Apply ingress yaml file and check ingress

```
root@Master:~# kubectl apply -f ingress.yaml
ingress.networking.k8s.io/nginx-ingress created
root@Master:~# kubectl get ingress
NAME            CLASS    HOSTS                ADDRESS    PORTS    AGE
nginx-ingress   nginx    teki.saber.com       80         8s
```

- Describe ingress nginx

```
root@Master:~# kubectl describe ingress nginx-ingress
Name:          nginx-ingress
Labels:        <none>
Namespace:     default
Address:
Ingress Class: nginx
Default backend: <default>
Rules:
  Host          Path  Backends
  ----          -
  teki.saber.com /    nginx-service:80 (192.168.219.101:80,192.168.219.87:80)
Annotations:    <none>
Events:
  Type    Reason    Age    From                      Message
  ----    -
  Normal  Sync      23s    nginx-ingress-controller  Scheduled for sync
root@Master:~#
```

2. Upgrade k8s cluster from 1.26 to 1.28 Version

- Check nodes and kubeadm version

```
root@Master:~# kubectl get nodes
NAME        STATUS    ROLES    AGE    VERSION
master      Ready     control-plane  24d    v1.31.14
worknode1   Ready     <none>      24d    v1.31.14
worknode2   Ready     <none>      24d    v1.31.14
root@Master:~# kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"31", GitVersion:"v1.31.14", Git
Commit:"5e00b99bac504844579ec74886b6cc5c9611ca19", GitTreeState:"clean", BuildDa
te:"2025-11-11T20:23:36Z", GoVersion:"go1.24.9", Compiler:"gc", Platform:"linux/
amd64"}
```

- Check kubelete version

```
root@Master:~# kubelet --version
Kubernetes v1.31.14
root@Master:~# kubectl version
Client Version: v1.31.14
Kustomize Version: v5.4.2
Server Version: v1.31.14
```

- Upgrade kubeadm

```
root@Master:~# sudo kubeadm upgrade plan
[preflight] Running pre-flight checks.
[upgrade/config] Reading configuration from the cluster...
[upgrade/config] FYI: You can look at this config file with 'kubectl -n kube-sys
tem get cm kubeadm-config -o yaml'
[upgrade] Running cluster health checks
[upgrade] Fetching available versions to upgrade to
[upgrade/versions] Cluster version: 1.31.14
[upgrade/versions] kubeadm version: v1.31.14
I0719 10:21:53.285608 15987 version.go:261] remote version is much newer: v1.3
6.2; falling back to: stable-1.31
[upgrade/versions] Target version: v1.31.14
[upgrade/versions] Latest version in the v1.31 series: v1.31.14
```

- Unhold kubeadm

```
root@Master:~# sudo apt-mark unhold kubeadm
Canceled hold on kubeadm.
root@Master:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
[137 kB]
```

- Now install kubeadm version

```
root@Master:~# sudo apt-get install -y kubeadm=<target-version>
-bash: syntax error near unexpected token `newline'
root@Master:~# sudo apt-mark hold kubeadm
kubeadm set on hold.
root@Master:~# kubectl drain master --ignore-daemonsets --delete-emptydir-data -
-force
node/master already cordoned
```

- Uncordon master

```
root@Master:~# kubectl uncordon master
node/master uncordoned
root@Master:~# apt-cache madison kubeadm
kubeadm | 1.31.14-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.13-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.12-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.11-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.10-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.9-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.8-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.7-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
```

- Now again upgrade kubeadm

```
root@Master:~# sudo kubeadm upgrade plan
[preflight] Running pre-flight checks.
[upgrade/config] Reading configuration from the cluster...
[upgrade/config] FYI: You can look at this config file with 'kubectl -n kube-sy
tem get cm kubeadm-config -o yaml'
[upgrade] Running cluster health checks
[upgrade] Fetching available versions to upgrade to
[upgrade/version] Cluster version: 1.31.14
```

```
root@Master:~# sudo apt-get install -y kubeadm=1.31.14-1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
kubeadm is already the newest version (1.31.14-1.1).
Solving dependencies... Done
0 upgraded, 0 newly installed, 0 to remove and 62 not upgraded.
```

- We can verify kubeadm updated

```
node/master uncordoned
root@Master:~# kubectl get nodes
NAME          STATUS    ROLES          AGE   VERSION
master        Ready     control-plane   25d   v1.31.14
worknode1     Ready     <none>          24d   v1.31.14
worknode2     Ready     <none>          24d   v1.31.14
root@Master:~# Connection to ec2-54-173-178-33.compute-1.amazonaws.com closed by
remote host.
```